

fine-tuned on the training dataset using sparse categorical cross-entropy five times with different random initialization seeds. To train the target MobileNetV2-based model, an ensemble of five obtained EfficientNet-B0 networks was used as a source for pseudo labels. The target model was trained using the KL-divergence as the distillation loss function, the distillation process is illustrated in Fig. 7. Adam optimizer was used for training both teacher and student models. To mitigate the problem of class imbalance, the authors used class weights (calculated from the proportion of samples per class) when calculating the cross-entropy loss for training teacher models. Stochastic weight averaging (SWA) was additionally applied to utilize the previously updated parameters from earlier epochs, which induces an ensembling effect [26]. The student model was trained for 110 epochs with early stopping.

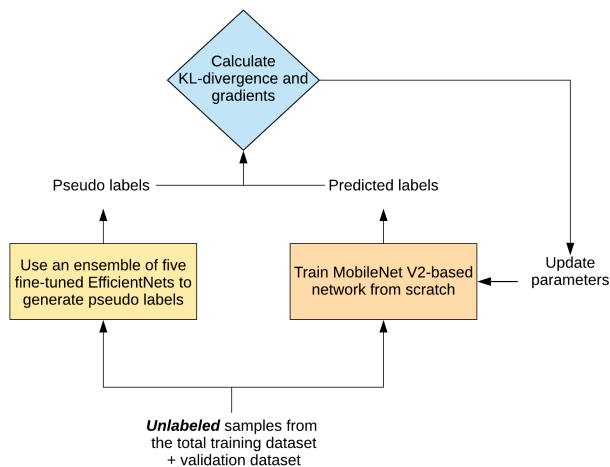


Figure 7. Knowledge distillation approach used by PyImageSearch.

4.8. Sidiki

This is the only solution that is not using any backbone networks. The proposed model can achieve an accuracy of over 80% on the validation set with only 60K parameters and the total model size of 73 KB. The model uses only 8 convolutional layers, 4 pooling layers and a fully connected layer. To speed up the inference, downsampling is applied in the first two convolutional layers by using strides 4 and 2 together with a small (6 and 12) number of channels.

The model was trained using Adam optimizer with a batch size of 32 as follows: for 10 epochs with a learning rate of $1e-3$, for 40 epochs with a learning rate of $1e-4$, then for 100 epochs with a learning rate of $7e-5$ and another 100 epochs with a learning rate of $9e-5$. After that, all layers of the model were prepared for Quantization Aware Training and then the model was trained for an-

other 100 epochs with a learning rate of $1e-4$, and for 100 epochs — with $1e-5$. Once the final accuracy of 80% was established on the validation set, the validation set was used for training of the quantized aware model for 50 epochs with a learning rate of $1e-4$. Then, the model was trained again on the original training set for 10 epochs with a learning rate of $1e-4$, and for 10 — with $1e-6$. Finally, the model was repeatedly trained around 6-8 times on the training and validation sets for 10 and 1 epoch, respectively.

5. Additional Literature

An overview of the past challenges on mobile-related tasks together with the proposed solutions can be found in the following papers:

- Learned End-to-End ISP: [20, 24]
- Perceptual Image Enhancement: [23, 17]
- Image Super-Resolution: [23, 34, 4, 41]
- Bokeh Effect Rendering: [15, 22]
- Image Denoising: [1, 2]

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ByteScene

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